Nowcasting

Workbook

Workshop 07: Pacific Islands Time-Series Database: Representation and Excel Workbook



IMF Institute for Capacity Development

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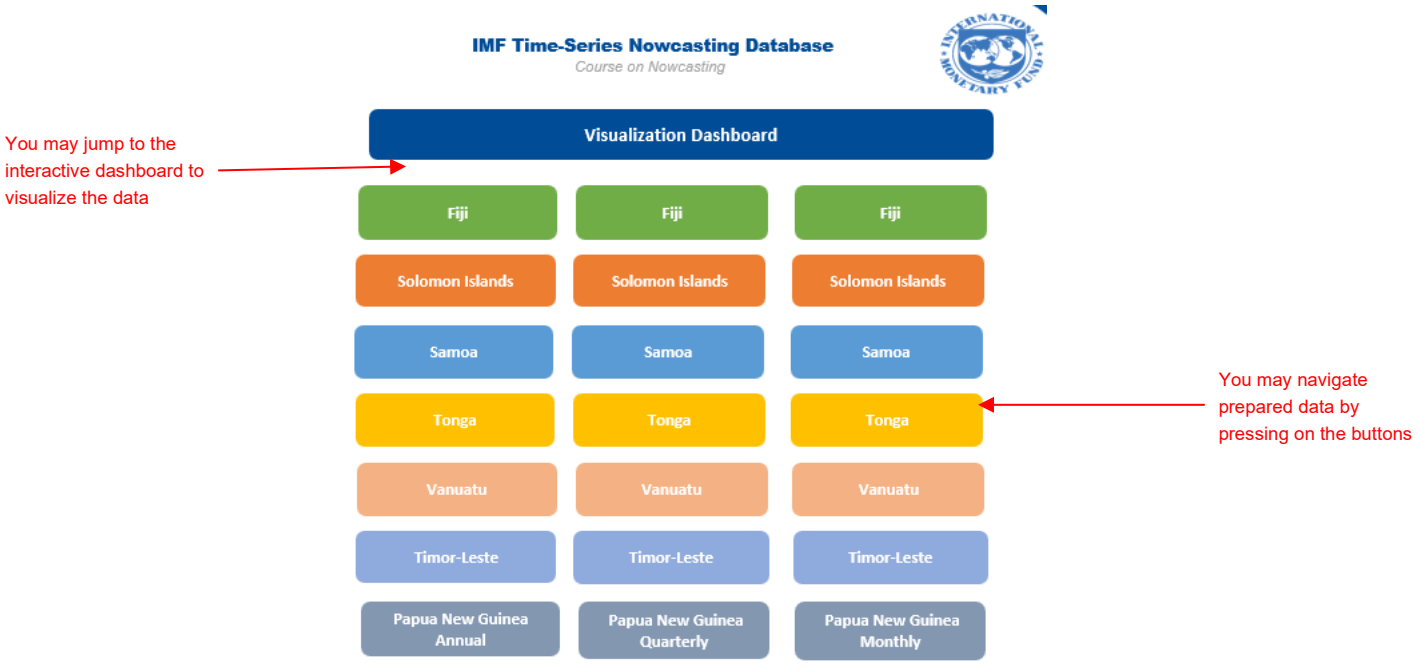
In prior sessions we covered the theoretical and empirical approaches of Nowcasting (i.e., Bridge, MIDAS, U-MIDAS) and we used data for India to illustrate the methods in the workshop sessions. You have been introduced to the tools and knowledge to produce your own nowcasting project. To this end, we have prepared a more comprehensive database to help you prepare the final participant presentation. You are tasked with applying any of the nowcasting methods to macroeconomic data of your choice and to prepare the PowerPoint presentation of your findings. This document provides a brief guide on using the database and the interactive dashboard. Please contact Ms. Anji Yang (ayang2@imf.org) if you have any question regarding this workbook.

1. WORKBOOK STRUCTURE

This workbook is divided into three sections: 1) Visualization Dashboard; 2) Data

The first contains a dashboard to easily visualize different indicators in annual frequency. The second section consists of selected high-frequency indicators for **Fiji, Papua New Guinea, Solomon Islands, Samoa, Timor-Leste, Tonga, and Vanuatu.** The dashboard and country data worksheets can all be accessed from the menu contained in the first worksheet "Guide".

Sections 2 and 3 below provide a detailed description of the workbook contents.



2. DATA DESCRIPTION

We prepared various indicators with Annual, Quarterly, and Monthly frequency in the Excel workbook. Please note that the datasheet does not encompass all the possible indicators available for the

nowcasting project. It is an incomplete list provided for your reference. Feel free to download other series based on your experience and judgement of what might influence the nowcast.

We have assigned a category for each series to better classify the data series. Categories used in the workbook include[1]:

- ***Target*:** Real Gross Domestic Product.
- ***Firm and Production:*** Electricity Consumption, Freight Traffic, Industrial Value Added, Inventory, Industrial Production, Motor Vehicle Output, Chemical Fertilizers Output, Fixed Asset Investment, and Real Estate Construction.
- ***Consumption*:** Retail Sales
- ***Financial Variables*:** Social Investment Financing, Stock Index
- ***Foreign Trade*:** Import, Export, and Trade balance, exchange rate, foreign reserves
- ***Prices*:** Consumer Price Index
- ***Balance of Payments:*** *Current/Financial/Capital Account, Balance on Goods and Services*
- ***Monetary Variables***: Central Bank Variables and Money Supply

All the data were collected from Haver Analytics and updated in July, 2026.

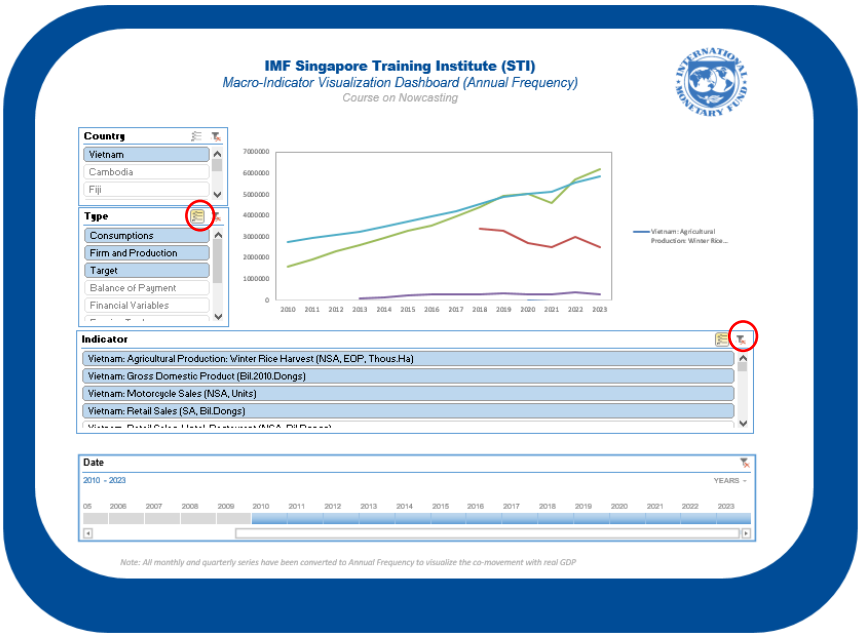
[1] Based on the High-Frequency Indicator category proposed by: Dauphin F Jean, et al. "Nowcasting GDP: A Scalable Approach Using DFM, Machine Learning and Novel Data, Applied to European Economies," IMF Working Paper 2022/052 (Washington: International Monetary Fund).

3. DASHBOARD

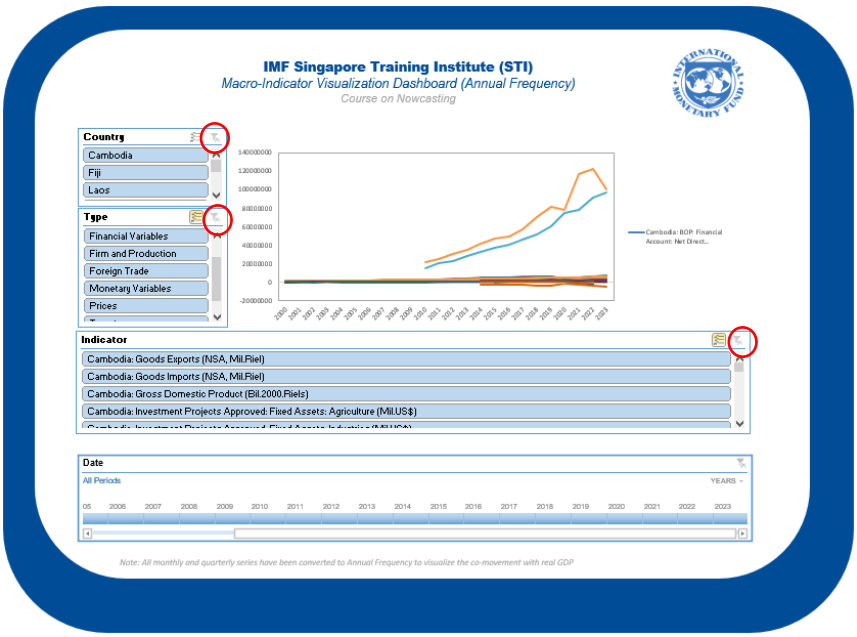
As mentioned previously, you may use the “Dashboard” tab to visualize any series of interest. Since the dashboard was designed to be interactive, you may select the category, descriptions, and time to customize the visualization chart.

: **Multi-select option**. Once you turn on this button, you may select more than one item from the list.

: **Clear all options.** Once you turn on this button, all your previous selection will be cleared. Note that you need to clear ALL the slicers to display all the options (starting point). An example is shown below:



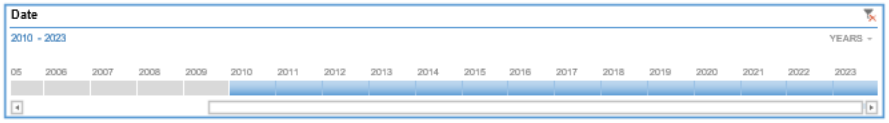
After you clicked all the “Clear all options” button, you will see all the available options appear (see below). Then, you can restart your selection of indicators to visualize.



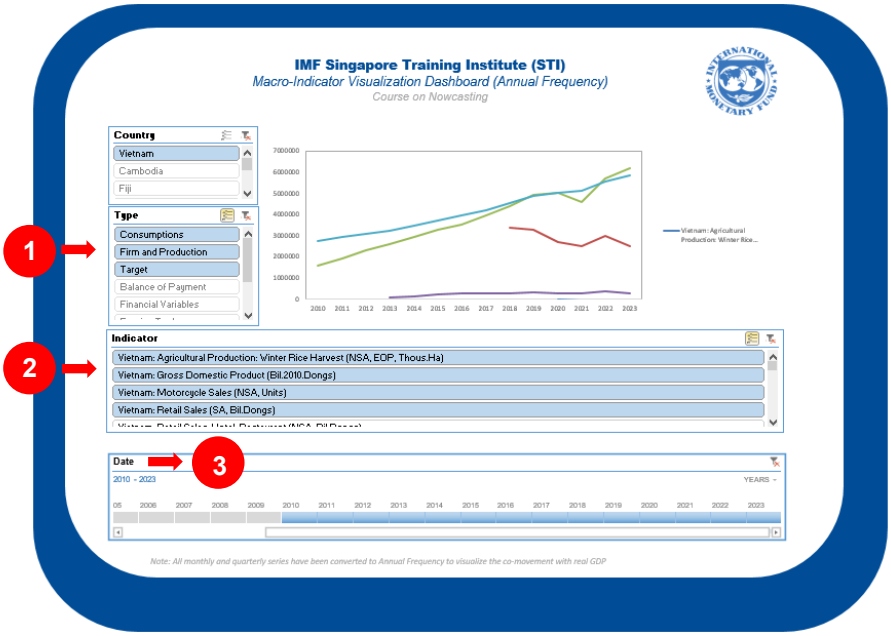
1) **Type**: you may choose one or more categories for next step.
2) **Indicator:** once you select the category, the indicator section will automatically present indicators that belong to that category for you to choose. Like other Excel charts, you may also change the format of the chart using the “Design” => “Change Chart Type.”



3) **Timeline:** After visualizing the indicator, you may also use the timeline to adjust the chart’s x-axis. In the timeline, blue means selected period, whereas gray indicates the unselected time period. You may modify the period by dragging the edge of the blue box.
Note*: the frequency drop-down menu of the timeline is only for your reference in better selecting the interest period. It will NOT convert the frequency of the data.*



The screenshot below illustrates the set-up options described above, using the Dashboard.



4. MINI EXERCISES TO FAMILIAR WITH THE WORKBOOK

- **Challenge # 1**: visualize Fiji's Real GDP and all indicators under "Consumptions" category, for 2010 ~ 2023 only.
- **Challenge # 2**: visualize Fiji's Real GDP and "International Visitor", for 2000 ~ 2023.

5. SOME TIPS IN CHOOSING THE INDICATORS FOR THE BASELINE MODEL

1) **Frequency:** the frequency of the selected indicators should be the same or higher than the target frequency (e.g., in the Vietnam workshops, the target variable is in Quarterly frequency, so our selected indicators should only be in Quarterly or higher frequency).

2) **Availability:** The selected indicators should have available data after the latest publication of the target variable.

3) **Correlations:** selected indicators should correlate with the target variable, in log terms for non-stationary indicators.

4) **Coefficients:** the sign of the estimated coefficients should make economic sense (match the economic priors).

5) **Recommended 5-Step Procedure in selecting the indicators:**

   i. Research the country's economy to understand the main drivers of economic growth in both expenditure and production sides (e.g., from GDP growth decomposition tables).

   ii. Collect data relevant to the country's economic growth.

iii. Filter the data using frequency, availability, correlations, and coefficients criteria.

iv. Use the selected data from step (iii) as the RHS and target variable as the LHS in the OLS regression. Drop indicators with a counter-intuitive sign, until all the explanatory variables' coefficients in the baseline regression make economic sense.

v. If step (iv) generates more than one candidate for the baseline model, use alternative regression selection metrics, such as Adjusted R-Squared and AIC criteria, to select the "best" model.